

Pre-Analysis Plan*

Power, bias, and the reproducibility of meta-analysis of psychology research

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1. Introduction

The aim of this paper is to assess the credibility of psychological research and the contributions of meta-analysis to psychological knowledge. We propose to survey 100 meta-analyses published by *Psychological Bulletin*. This survey will focus on three critical dimensions of the credibility of psychological research: reproducibility, statistical power, and bias.

How powerful is the typical psychological study? How replicable and robust are the central findings of psychological meta-analyses? What is the relative size of the typical bias due to selective reporting (aka, publication bias, small-sample biases and *p*-hacking)?

Statistical power is a critical element of credible research (Cohen, 1988). The absence of power is likely to be a principal cause of the current replicability crisis (Nosek *et al.*, 2015). Hence, our first objective is to explore whether there is sufficient power among psychological studies surveyed by these 100 meta-analyses to say anything meaningful about the psychological phenomenon in question.

How robust and replicable is meta-analysis? Replication is the hallmark of science (Popper, 1959). Similarly, the replication of meta-analyses is critical for the credibility of psychological science. Hence, our second objective is to re-analyze these 100 meta-analysis to gauge how replicable and robust the central findings from psychological meta-analyses are.

Our third objective is to calculate a lower bound of the magnitude of likely selective reporting bias in each area of psychological research represented by these 100 meta-analyses. For decades, researchers from many disciplines have recognized the importance of ‘publication bias’; that is, the selective reporting of statistically significant results and *p*-hacking (Sterling, 1959; Rosenthal, 1979; Hedges and Olkin, 1985; Begg and Berlin, 1988; Sterling, Rosenbaum and Weinkam, 1995; Copas, 1999; Senn, 2008; Stanley, 2008; Mandel and Rinott, 2009; Rücker *et al.*, 2011; Simonsohn, *et al.* 2014; and Stanley and Doucouliagos, 2014 to mention a few). Low power and selective reporting bias are largely responsible for the current ‘credibility’ or ‘replication crisis’ in psychology. We intend to gauge their scope and magnitudes across these 100 areas of psychological research.

* Adapted from Alejandro Ganemian's template for pre-analysis plans, <http://www.bitss.org/resource-tag/pre-analysis-plans/>; accessed 5/26/2016. Because we are essentially conducting a meta-meta-analysis, pre-analysis issues suitable for primary experimental or observational research are largely irrelevant. Nonetheless, we follow the spirit of Alejandro Ganemian's template for pre-analysis plans if not its letter.

2. Research Strategy

2.1 Sampling

2.1.1 Sampling Frame

We will re-analyze the 100 most recently published (as of June 1, 2016) meta-analyses in the *Psychological Bulletin*. Thus, we need merely to obtain the research data used by these 100 meta-analytic studies. Most of this ‘meta-data’ will be posted on various repositories and websites, or in the published studies. When the full dataset is not publically posted, we will contact the authors of the meta-study and request that they share their data. Willingness to share data is one dimension of replicability.

Our ideal population is all empirical psychological research. However, our sampling frame is that research meta-analyzed by papers in the *Psychological Bulletin*. From this sampling frame our convenience sample will be the 100 most recently published meta-analyses in the *Psychological Bulletin*. To the extent that these meta-analyses are representative of population of all empirical psychological research, our findings will also be representative of psychological research, in general. However, if these meta-analyses are not representative of psychological research, our study results will not be generalizable to psychological research, in general, but merely to those represented by meta-analyses in the *Psychological Bulletin*. *Psychological Bulletin* is the paramount publication outlet for meta-analyses in psychology. Because it is a top APA journal, the meta-analyses published there are, at a minimum, representative of the psychological research that APA values. Nonetheless, we will explicitly and clearly state these limitations of generalizability in any resulting paper or report.

Our reason for taking this convenience sample is that our project is a response to the *Psychological Bulletin*’s “Call for Papers: Replication and Reproducibility”— see [APA’s website](#). We choose the most recently published meta-analyses because they are likely to be more representative of current psychological research, and our sample size of 100 areas of research is sufficiently large to generalize to our sampling frame (the population of meta-analyses published in the *Psychological Bulletin*), if not to our ideal population of all empirical psychological research. Our research plan has been provisionally approved for a special issue of the *Psychological Bulletin* on “Replication and Reproducibility.”

2.1.2 Power

Statistical power is not a relevant issue for our survey as it would be for primary research. We will report descriptive statistics concerning the median power and bias of these 100 areas of psychological research. That is, we will make thousands of *ex post* calculations of statistical power of individual psychological studies using the weighted average of all effects sizes (fixed-effect or equivalently the unrestricted weighted least squares) as the proxy for ‘true effect.’ Because we are not reporting *p*-values or conducting hypothesis tests on these magnitudes but merely presenting descriptive statistics, the statistical power of our ‘analysis’ is not relevant.

2.2.2 Data Collection

As already stated, we will collect data from: published meta-analysis papers, various repositories, and from supplementary material associated with these meta-analyses. When the meta-data are not readily available, we will contact the authors directly and request their data.

From each of the 100 meta-studies, we will collect, or code, comparable effect sizes and their estimated standard errors. From this core data we will be able to calculate statistical power, identify and correct publication bias, and replicate the core meta-analysis findings.

2.2.2 Data Processing

Data processing may take up to six months because the data will likely come in only gradually. Each of the meta-analyses will need to be recalculated and potential selective reporting bias will be reduced using several recent methods (Stanley and Doucouliagos, 2014; Ioannidis et al. 2016). We will also calculate median power, the percent of studies with adequate power, heterogeneity among studies, and average research inflation for each of these 100 meta-analyses.

There is no need to keep the raw data confidential because it is all public already. Meta-analysis data contains no information on individual subjects. Nonetheless, we will not place copies of this meta-analysis data on the internet nor on any publically accessible device or service. Nor will we share the raw meta-analysis data with anyone outside our research team. We will be the sole ‘owners’ of the data. The aggregate data that we calculate with identifiers for individual meta-studies removed will be made publically available through the Open Science Framework at the end of the project.

3. Empirical Analysis

3.1 Variables

The variables that we wish to report descriptively concern statistical power, heterogeneity, and average selective reporting bias (aka publication bias).

To calculate power, we use a conventional 5% level for statistical significance and, following Cohen (1988), the conventional 80% level as the benchmark for adequate power. Given these long accepted conventional values, it is easy to show that an estimate has adequate power when its standard error is smaller than the absolute value of the ‘true’ underlying effect divided by 2.8. We will use this cut-off to assess whether each of the thousands of individual studies among these 100 meta-analyses have adequate power. As recommended by Ioannidis *et al.* (2016), we will take a conservative approach to calculating power by using the fixed-effect weighted average (FE or its equivalent WLS weighted average) of all reported effect sizes as the proxy for the ‘true’ effect size. Although FE is less biased than random effects in the presence of selective reporting, FE is also known to exaggerate effect size when there is selective reporting (Stanley and Doucouliagos, 2014 & 2015). Thus, using FE as the ‘true’ effect will, if anything, overestimate statistical power.

Measures of heterogeneity (*e.g.*, Higgins’ H , and I^2) will also be coded and/or calculated for each of these 100 meta-analysis (Higgins and Thompson, 2002).

Lastly, we will gage the average magnitude of residual reporting bias for these 100 areas of psychological research. We quantify this magnitude as the difference between the average reported effect size and a less-biased average effect size, including: the weighted average of the adequately powered (WAAP) (Ioannidis et al., 2016) and alternative corrections for publication bias; for example, *PET-PEESE* (precision effect test-precision effect estimate with standard error) (Stanley and Doucouliagos, 2014). Because this project is largely the application of the approach that we took in a similar study of economics research, greater detail about these variables and related methods can be found in that paper, Ioannidis et al. (2016).

3.2 Outliers and erroneous meta-analysis

Because we will be re-calculating the central results of these 100 meta-analyses, we may come across some published meta-analyses with errors. That is, those meta-analyses that were calculated incorrectly or where the shared meta-data cannot be correct. When we find any indication of mistakes or erroneous meta-data, we will contact the authors to identify the source of the inconsistency. We will record and report such incidences as failures to replicate. If we cannot correct or explain the identified problem satisfactorily, we will remove that meta-analysis from our reported descriptive statistics.

4. Research Team

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Doucouliagos and Carter will collect the 100 meta-datasets. Stanley and Carter will do all of the necessary calculations of meta-analysis statistics, statistical power, heterogeneity and average selective reporting bias. All three of us will be involved in the interpretation of our findings and in the writing of the resulting paper.

5. Deliverables

The goal of this study is to produce a paper for publication in the *Psychological Bulletin* for a special issue on “Replication and Reproducibility.” The deadline for submission is June 30 2017. Stanley will be the lead author of this paper.

6. Calendar

Data collection will commence June 1, 2016, analysis, and data collection will continue for several months, to be completed by Dec 31, 2016. A draft of the paper will be written by Jan 31, 2017.

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